Fidetype: a vendor-neutral stream format and trust layer for safer neural text entry across MEG, intracortical and EMG decoders

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Abstract

Neural text-entry decoders, from non-invasive magnetoencephalography (MEG) to intracortical arrays and wearable electromyography (EMG), now produce fluent text. They rarely signal when that text is wrong, and each device reports confidence in its own format. We present Fidetype, an open, vendor-neutral stream format for per-frame decoder output (NeuralTextStream) and a decoder-agnostic trust layer that reads only this stream. The layer calibrates confidence; chooses between committing a sentence, offering choices and abstaining; bounds the expected wrong-commit rate with conformal risk control; keeps that bound under within-session drift by online re-calibration from user corrections; and records each decision in a hash-chained provenance ledger. We apply the same layer, unchanged, to three device types with public data: MEG typing (19 participants), intracortical speech (one participant) and forearm surface EMG key presses (19 participants). We report effective words per minute (eWPM) of a simulated user against a gaze-typing baseline, with shuffled-input and null-input controls. The layer reduced the wrong-commit rate from 0.610 to 0.059 on intracortical speech with no measurable speed cost, from 0.54–0.96 to at most 0.004 on EMG, and from 0.997 to 0 on MEG, and it committed no text from null input. Conformal risk control brought the mean wrong-commit rate to 0.046–0.048 (target 0.05). Under drift, online re-calibration held it at 0.048–0.049, where offline calibration gave 0.0625–0.0635. On MEG, where the decoder almost never proposed the right sentence, the layer committed nothing, and the system only reached parity with gaze typing (difference +0.05 words per minute, 95% CI over sentences $[-0.04, +0.15]$).

1 Introduction

Brain–computer interfaces (BCIs) for text entry are moving from the laboratory towards products. Intracortical speech neuroprostheses decode attempted speech at conversational rates [1, 2]; handwriting and typing neuroprostheses reach similar speeds [3, 4]; non-invasive decoders now reconstruct typed sentences from MEG [5, 6]; and wrist or forearm surface EMG (sEMG) is a candidate input for consumer devices [7, 8]. Most of these systems end in a language-model (LM) stage that turns noisy per-frame predictions into fluent sentences. This stage improves the error rate, but it also makes wrong output look right: a fluent wrong sentence is harder to notice than a garbled one, as known from automatic speech recognition [9]. For a user who depends on the interface to communicate, a wrong sentence that is committed and sent costs more than a slow one.

Three gaps stand between current decoders and safe use.

No common format for decoder output with honest confidence. Each decoder reports its output in its own way. The Apple BCI Human Interface Device (HID) profile sends discrete events with a 0–255 "neural activity strength" that is not a probability [10]. W3C EMMA can carry a confidence and an n-best list, but it has no per-frame distributions [11]. Lab Streaming

Layer and BIDS move or store raw signals, not decoded text [12]. A downstream component that wants to decide whether to trust a decoder must therefore be rebuilt for every device.

No decoder-agnostic rule for *when to commit*, with a guarantee. Calibration [13] and selective prediction [14] are well studied, and recent BCI work adds confidence estimates to single decoders [15]. We know of no layer that works across device types, decides between committing, offering alternatives and abstaining, and states a finite-sample bound on how often it commits wrong text. **[TODO: G10: add and position the closest work (Huang et al. 2025, co-control BCI; Damri & Laufer-Goldshtein 2026, risk-controlled generative ASR) after their source pages exist.]**

Error rates do not show whether a BCI saves the user time. Character and word error rates (CER, WER) compare decoders, but they do not say whether a user would type faster with the BCI than without it. A no-BCI baseline, such as gaze typing for people with ALS [16], is rarely reported next to the decoder.

This work. We address the three gaps with one format, one layer and one evaluation protocol, and we test them on three device types with public data (Figure 2). Our contributions are:

1. **A vendor-neutral stream format** (NeuralTextStream v0.1) for neural decoder output: per-frame log-probabilities over a declared vocabulary (characters, keys, phonemes), with timing, calibration and provenance conventions, a conformance suite of 60 checks and 52 golden test vectors, and adapters for MEG, intracortical and sEMG decoders. No breaking change was needed across the three device types (Section 3).

2. **A decoder-agnostic trust layer** that reads only NeuralTextStream: temperature calibration, a neural-evidence guard, and a COMMIT/OFFER/ABSTAIN policy fitted for maximum eWPM under a wrong-commit limit. It also works as a gate on a black-box, LM-rescored decoder (Section 4).

3. **Finite-sample and drift-robust wrong-commit control**: conformal risk control (CRC) and Learn-then-Test (LTT) for the commit threshold [17, 18], and an online controller that re-calibrates from user feedback, with a long-run bound that holds for any sequence of items for two of its variants [19, 20].

4. **A privacy-preserving provenance ledger**: an append-only, SHA-256 hash-chained record of every trust decision and user correction, without raw signals.

5. **A cross-device evaluation protocol**: eWPM of a simulated user against a no-BCI gaze-typing baseline, wrong-commit and silent-error rates, and shuffled, zero, permuted, frame-shuffled and other-sentence controls (Section 5).

6. **An open release** under Apache-2.0 of the specification, reference implementation, adapters and experiment scripts (Section 10).

Figure 1 summarises the main result. On all three device types the trust layer moved the system from frequent wrong commits to almost none. On intracortical speech and on session-calibrated sEMG, it did so at no speed cost and well above gaze typing. On MEG, it removed all wrong commits, but only by committing nothing, and the system only reached parity with gaze typing.

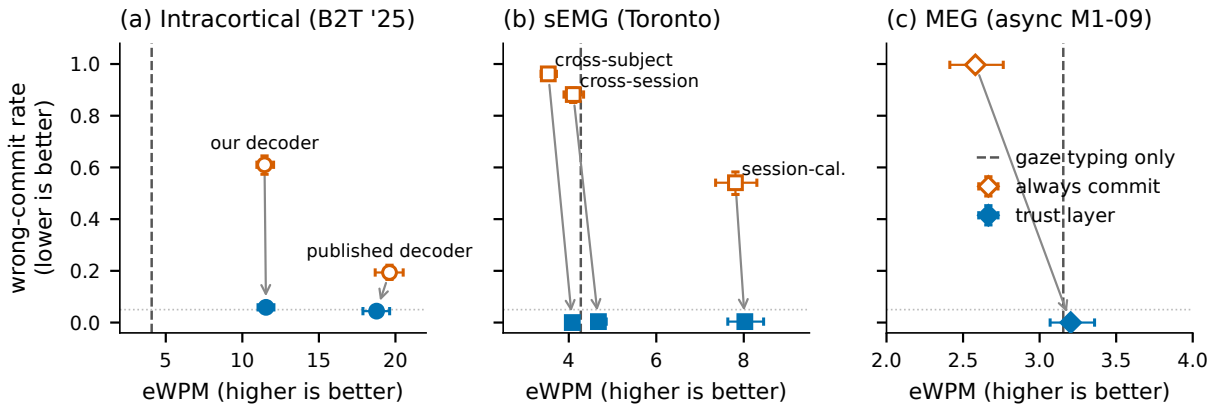


Figure 1: The same trust layer removes almost all wrong commits on three device types. Each panel plots the wrong-commit rate (share of sentences committed with an error) against effective words per minute (eWPM) of a simulated user. Open markers: always commit the decoder's top sentence. Filled markers: the trust layer on the same decoder output; arrows connect the two. Dashed line: gaze typing alone (no BCI). Dotted line: target $\alpha = 0.05$. Error bars: 95% bootstrap CIs over trials (a) and sentences (b, c; cluster bootstrap over the 17 unique sentences in c). (a) Intracortical speech, Brain-to-Text '25, one participant, 816 test trials; "our decoder" is our phoneme-to-word decoder, "published decoder" is the published LM-rescored sentence. (b) Forearm sEMG, 551 composed test sentences from 19 participants, three decoder conditions. (c) MEG typing, 637 test instances (17 unique sentences) from 19 participants, our asynchronous M1-09 decoder.

2 Related work

Brain-to-text decoders. Non-invasive MEG decoding of typed sentences reached a mean CER of 0.29 with a keystroke-locked (synchronous) model [5]. A second version decodes sentences from the continuous recording without key times (asynchronous) and adds an LLM stage [6]. Other non-invasive work decodes perceived speech [21] or words with large pre-trained models [22, 23]. Intracortical decoders reach high accuracy for attempted handwriting [3], speech [1, 2, 24] and typing [4]; public benchmarks expose their phoneme logits [25, 26]. Surface EMG typing datasets now exist at scale for the wrist [7] and the forearm [8]. All these decoders use CTC or a related sequence loss [27] with beam search and an LM [28]. None of them standardises its output, and none decides when its output is safe to commit.

Calibration, selective prediction and abstention. Modern neural networks are overconfident; temperature scaling is a simple and strong fix [13], but calibration degrades under dataset shift [29]. The maximum softmax probability is a strong baseline for detecting errors [30], and selective classification trades coverage for risk [14]. In BCI, confidence-aware decoding with abstention was shown for EEG speech commands on one device [15]. **[TODO: G10: add Huang et al. 2025 (calibrated alternatives for brain-to-text fusion) once a source page exists.]**

Conformal prediction and risk control. Conformal methods give distribution-free, finite-sample guarantees [31]. Conformal risk control (CRC) bounds the expected value of a monotone loss [17]; Learn-then-Test (LTT) bounds the risk with high probability by multiple testing [18]. Under distribution shift, weighted and non-exchangeable variants give approximate bounds [32–34], and online methods adapt a threshold from feedback with long-run guarantees for arbitrary sequences [19, 20, 35, 36]. Conformal methods were applied to speech recognition [37] and to LM outputs and abstention [38–40]. To our knowledge, they have not been applied to the commit decision of neural text entry, nor across device types.

Hallucination in LM-assisted decoding. LM stages can produce fluent text that the input does not support. In speech recognition this causes real harm [9], and generative error correction can replace correct words [41]. Reliability-oriented metrics that reward abstention on uncertain input are emerging for ASR [42].

Text-entry and BCI metrics. BCI work often reports information transfer rate [43] or CER/WER; text-entry research uses error metrics such as MSD and KSPC [44]. Studies with gaze-typing users quantify the time cost of selections and corrections [16]. We combine these into an effective-words-per-minute measure with an explicit no-BCI baseline.

Interoperability standards and benchmarks. Apple's BCI HID profile [10], W3C EMMA [11], Lab Streaming Layer [12], BIDS, and IEEE P2731 terminology [45] each cover part of the stack: OS input events, multimodal annotation, raw-signal transport, data at rest and vocabulary. Benchmarks such as MOABB and NeuralBench unify decoder evaluation [46, 47]. None of them defines a per-frame decoded-output format with confidence semantics, which is the gap NeuralTextStream fills. **[TODO: add a BIDS citation (Gorgolewski et al. 2016) after it is read.]**

3 The NeuralTextStream format

NeuralTextStream (NTS) is the contract between a decoder and any consumer of its output (Figure 2). A consumer does not need to know how the decoder works. The full specification, v0.1, is in the repository (`spec/neuraltextstream-v0.1.md`); here we summarise its design.

Frame core. A stream of T frames over a vocabulary of V tokens holds a $T \times V$ array `log_probs` of natural-log probabilities (row t is a distribution at frame t), the frame times in seconds, the ordered vocabulary `vocab`, its type `vocab_type` (`char`, `key`, `word`, `phoneme`, `command` or `other`) and `blank_id`, the index of the CTC blank or `null` for non-CTC decoders. Optional fields give the nominal `frame_period`, the algorithmic `latency` (look-ahead in seconds; 0 means causal), the producer name and a `meta` object. A stream may have $T = 0$.

Semantic conventions. Following the OpenTelemetry approach of named attribute namespaces, `meta` keys use defined prefixes: `device.*` (for example `device.modality`), `decoder.*` (name, version, `decoder.causal`, `decoder.weights_sha256`), `lm.*`, `adapter.*`, `calibration.*`, `data.derived` and `language`. Each field has a stability level; v0.1 is a Development release.

Honest probabilities. A producer must emit the decoder's own output and must not sharpen or smooth the probabilities for display; calibration is the consumer's job. A number is a probability only if the stream says so. Ground-truth text must not appear in the header or `meta` (a leakage rule), and raw signals and direct identifiers stay out of the stream (privacy rules).

Serialisations and conformance. The reference serialisations are an NPZ file and a JSON Lines streaming form; WebSocket, LSL and HID bridges are sketched. A conformance suite (our code) runs 60 checks, each tied to a specification clause, and reports one of three cumulative levels: *core* (a consumer must accept the stream), *trust-ready* (a trust layer can use it safely: modality, causality, calibration state and language are declared) and *provenance* (a decision can be traced to the producing model: stream id, decoder name, version and weights hash). It ships 52 golden test vectors (32 NPZ, 20 JSON Lines); the reference consumer passes 52 of 52. On synthetic inputs, our MEG and sEMG adapters reach the provenance level and the intracortical

Table 1: Devices, data and streams. One adapter per device type maps decoder output into NeuralTextStream. All datasets are public. Test units are those of the trust evaluation.

	MEG (typing)	**Intracortical (speech)**	**Forearm sEMG (keys)**
Dataset (licence)	SpanishBCBL (CC BY-NC 4.0)	B2T '25 copy task (CC0)	Toronto sEMG (CC BY 4.0)
Participants	19	1 (T15)	19
Test unit	637 instances, 17 unique sentences	816 trials, 14 days	551 composed sentences
Decoder	own async ConvCTC (medium)	published online RNN; own phoneme-to-word route	own CNN
NTS `vocab_type`	`char` (CTC)	`phoneme` (CTC, 41 tokens)	`key` (W3C codes)
Frame	one per MEG frame	0.08 s	one per key press

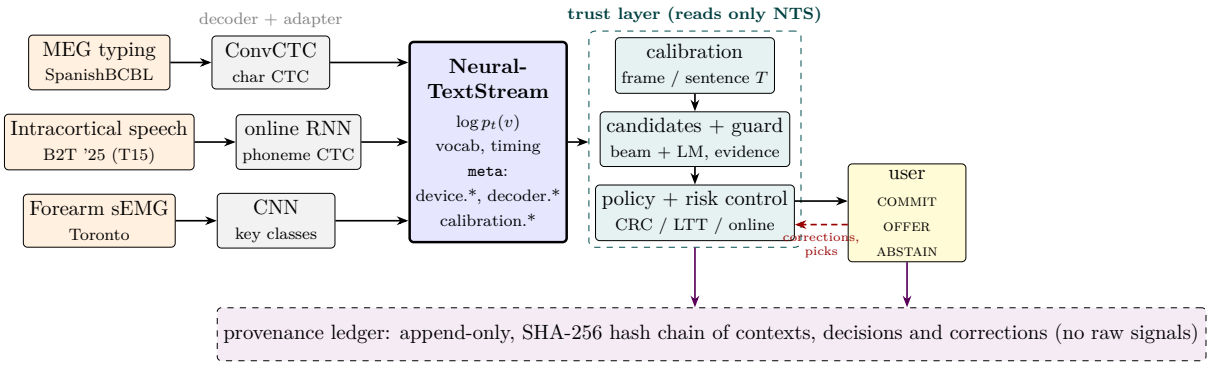


Figure 2: Architecture. Each device's decoder emits a NeuralTextStream through a small adapter. The trust layer reads only the stream: it calibrates confidence, builds candidate sentences with a neural-evidence guard, and chooses an action under a risk-control rule. The user's corrections and picks (dashed) feed the online re-calibration. Every context, decision and correction goes to an append-only, hash-chained provenance ledger that stores no raw signals.

adapter reaches trust-ready, because the published online decoder has no weights file to hash. Writing the suite found five bugs in our own producers and ten ambiguities in the specification text, now listed as open questions.

Three device types, no breaking change. Table 1 lists the three adapters. The MEG decoder emits character CTC frames; the intracortical stream carries the published online phoneme CTC logits (41 tokens, 0.08 s frames); the sEMG stream has one frame per key press over W3C key codes with `blank_id = null`. Mapping the second and third device types into NTS needed no change to the frame core. The trust harness needed one optional field (the truth in the stream's token space, because phoneme truth differs from the word truth that the user sees), and provenance added optional, non-breaking `meta` keys.

4 The trust layer

The trust layer reads one NeuralTextStream per sentence attempt and returns one action. It never reads raw signals or decoder internals. Figure 3 shows the decision flow.

4.1 Calibration

Decoder log-probabilities are often over-confident. We fit a frame temperature T_f that rescales every frame, $\tilde{p}_t(v) \propto p_t(v)^{1/T_f}$, by minimising the negative log-likelihood of the truth on fit data, and a sentence temperature T_s for the sentence-level confidence $\hat{p}_{\text{top1}}$, the normalised probability

of the top candidate among the n-best [13]. We report the expected calibration error (ECE) of $\hat{p}_{\text{top1}}$ before and after.

4.2 Candidates and neural-evidence guard

Candidate sentences come from a CTC prefix beam search with a character 6-gram LM and a word unigram, both trained on Tatoeba sentences, and an n-best list (n = 10) [27, 28]. For the published intracortical sentences, the candidate is the given text. Because the LM can write fluent text from any input, a guard scores how much of each candidate the LM invented. It combines two signals: the share of the candidate's characters whose posterior under the calibrated frames alone (no LM) is low (*low evidence*), and one minus the posterior-weighted word agreement between the top candidate and the other n-best candidates (*n-best inconsistency*); the guard risk is the maximum of the two weighted signals. A candidate with high guard risk cannot be committed.

4.3 Decision policy

The policy chooses one of four actions: COMMIT the top candidate; OFFER the top k candidates for the user to pick; OFFER_WORDS, an offer with uncertain words highlighted; or ABSTAIN, after which the user types the sentence with the fallback keyboard. The thresholds (commit probability, guard risk, offer size, abstain risk, fallback) are chosen on fit data from a small grid to maximise eWPM (Section 5) subject to a wrong-commit rate ≤ 0.05 on the fit data. This plug-in fit has no guarantee on new data, which motivates Section 4.4.

4.4 Finite-sample risk control for the commit threshold

Let s_i be the score of item i ($\hat{p}_{\text{top1}}$ if the other policy rules allow a commit, else $-\infty$) and λ the commit threshold. The loss is $L_i(\lambda) = c_i$ if $s_i \geq \lambda$ (with $c_i = 1$ if the top candidate is wrong) and f_i otherwise (the loss of the non-commit action, 0 here), made monotone in λ by taking $\max(c_i, f_i)$. On n calibration items with known truth, conformal risk control [17] chooses

$$\hat{\lambda}_{\text{CRC}} = \inf\Big\{\lambda : \tfrac{n}{n+1}\hat{R}_n(\lambda) + \tfrac{1}{n+1} \leq \alpha\Big\}, \tag{1}$$

where $\hat{R}_n$ is the empirical risk. If calibration and test items are exchangeable, $\mathbb{E}[L_{n+1}(\hat{\lambda}_{\text{CRC}})] \leq \alpha$. No feasible λ exists for $n < 1/\alpha - 1$ (19 for $\alpha = 0.05$); the layer then never commits. Learn-then-Test [18] instead tests H_0: $R(\lambda) > \alpha$ along a decreasing grid with exact binomial p-values and fixed-sequence testing, which gives $\Pr(R(\hat{\lambda}_{\text{LTT}}) \leq \alpha) \geq 1 - \delta$. Any threshold above $\hat{\lambda}$ keeps the guarantee, so the layer can combine CRC with the eWPM-tuned threshold as $\hat{\lambda} = \max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}})$. The other parts of the pipeline (temperatures, guard, other thresholds) are fitted on separate data. We use $\alpha = 0.05$ and $\delta = 0.1$ throughout.

4.5 Online re-calibration under drift

Exchangeability fails when a session drifts. The online controller re-fits the threshold during the session from user feedback (Figure 4). Which feedback exists depends on the action: a corrected commit reveals a wrong commit; an uncorrected commit is right only if silence means right; an offer reveals the truth when the user picks or retypes; an abstention reveals nothing unless the final text is logged. The controller therefore accepts two feedback levels: the *realised loss* of committed items, and, when the final text is known, the counterfactual commit loss of non-committed items. Feedback can arrive late and out of order. Table 2 lists the four methods.

The `aci` rule follows adaptive conformal inference and online risk control [19, 20, 36]; `aci_window` applies the ACI update to the level of a windowed CRC [19]. We use $N = 200$, $\rho = 0.99$, $\eta = 0.05$ and $\gamma = 0.005$, fixed before the evaluation. These bounds are long-run

Table 2: Online re-calibration methods for the commit threshold. L_t: realised loss at item t; T: number of items in the session. The two `aci` variants bound the long-run average loss for any sequence of items, without exchangeability.

Method	Update	Guarantee	Feedback
`window`	CRC on the last N items with known truth	approximate	full
`decay`	weighted CRC, weights ρ^{age} [33, 34]	approximate	full
`aci`	$\lambda_{t+1} = \max(\lambda_{\min}, \lambda_t + \eta(L_t - \alpha))$	$\frac{1}{T}\sum L_t \le \alpha + \frac{\lambda_{T+1}-\lambda_1}{\eta T}$	realised only
`aci_window`	window CRC at level α_t; $\alpha_{t+1} = \alpha_t + \gamma(\alpha - L_t)$	$\frac{1}{T}\sum L_t \le \alpha + \frac{\alpha_1-\alpha_{T+1}}{\gamma T}$	full + realised

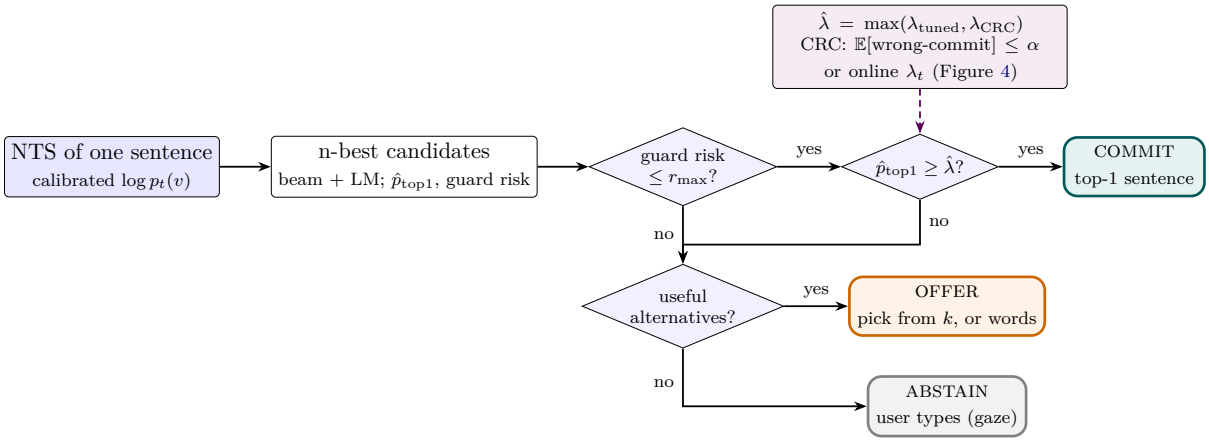


Figure 3: Trust decision for one sentence. Candidates come from the calibrated stream. A candidate is committed only if the neural-evidence guard accepts it and its calibrated confidence reaches the threshold $\hat{\lambda}$, which is the larger of the eWPM-tuned threshold and the CRC threshold (or the online threshold λ_t). Otherwise the layer offers alternatives or abstains, and the user types with the fallback keyboard.

averages over the T items of a session; they say nothing about short windows, and they cover only the errors that the system observes. A wrong commit that the user does not notice (a *silent error*) is outside them.

4.6 Provenance ledger

Each decision is appended to a JSON Lines ledger. Entry 0 is a genesis record; *context* entries are content-addressed (their id is the SHA-256 of the context) and hold the adapter, decoder (name, version, weights hash, training data and licence), LM, calibration (temperatures, risk-control method, α, δ, threshold, n) and policy versions; *decision* entries hold the action, scores, a hash of the NeuralTextStream and its frame range; *correction* entries link to the decision they correct and store the edit distance. Each entry stores `prev_hash` and its own SHA-256 over the canonical JSON, so any edit, deletion or reordering breaks the chain; a separately stored head anchor detects truncation. The ledger stores no raw signals (numeric arrays longer than 64 values are rejected), stores text as a keyed hash by default, and pseudonymises subject and session ids. It is tamper-evident, not tamper-proof: there is no signature in v0.

5 Evaluation protocol

5.1 Effective words per minute and error measures

Error rates compare decoders but do not tell whether a user saves time. We therefore simulate a user who reads the system output, accepts or corrects a commit, picks from an offer, or types

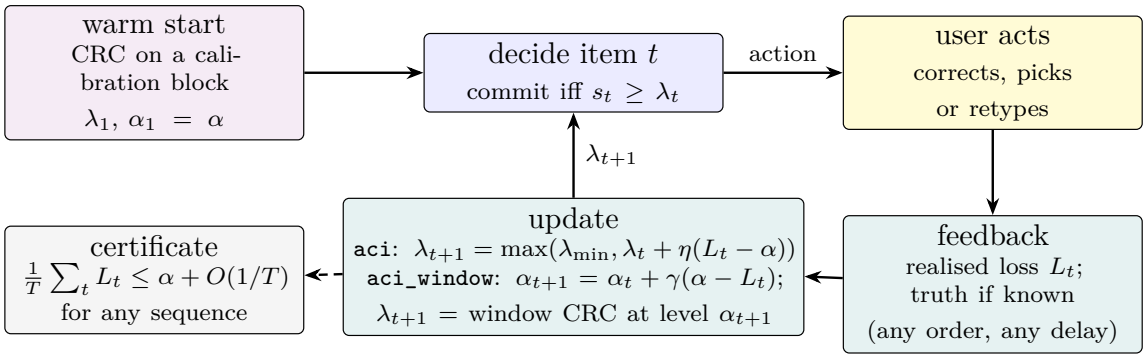


Figure 4: Online re-calibration loop. A short calibration block gives the starting threshold. After each item, the realised loss (and the truth, if the final text is known) updates the threshold. For `aci` and `aci_window`, the running wrong-commit rate is bounded by α plus a term that shrinks as $1/T$, for any sequence of items.

the sentence with a fallback keyboard, and we report the *effective words per minute* (eWPM): correct words in the final text divided by the total simulated time, over all test items. The *wrong-commit rate* is the share of items for which the system committed a sentence that differs from the truth (for OFFER_WORDS, a sentence with a wrong word that was not highlighted). A *silent error* is a wrong commit that the simulated user does not notice; it reaches the recipient.

5.2 User model and baseline

The user model (`UserConfig.aac_gaze()`) represents a gaze-typing user. The fallback keyboard runs at 20 characters per minute, and the time to select one offered slot sums to 4.5 s; both are matched to measurements of eye-gaze typing users with ALS [16]. The neural input runs at a device-specific speed (B2T '25: the measured 159.3 characters per minute; sEMG: 75.8 characters per minute, from the 0.792 s press period; MEG: 152 characters per minute). A wrong commit without a flag is missed with probability `p_miss_unflagged` = 0.05. Other time costs (reading per word and per candidate, correction overhead) are literature-based assumptions listed in Appendix A. The baseline is *gaze typing only*: the same user types every sentence with the fallback keyboard and no BCI. **[TODO: G6: eWPM sensitivity to the user model on all three devices (fallback 10/20/40 characters per minute, selection time $\times 0.5/\times 2$, `p_miss_unflagged` 0/0.05/0.2).]**

5.3 Datasets and splits

MEG typing (SpanishBCBL). Healthy participants typed memorised Spanish sentences in an MEG scanner [5, 48]. We use 19 unique MEG participants and split by a hash of the sentence text, so no test sentence appears in training or in the LM text. The test set has 637 instances of 17 unique sentences. Our asynchronous ConvCTC decoder (4 Conformer layers, model width 256 [49]) receives the MEG segment from 0.4 s before the first key press to 0.5 s after the last, its length and the participant index, but no key times at inference; it uses key times as auxiliary training labels (Appendix B). All MEG trust results use the streams of this asynchronous decoder; a synchronous decoder with oracle key timing (M1-15) gives the same picture and is reported in Appendix B.

Intracortical speech (Brain-to-Text '25). The public copy-task file holds the online phoneme CTC logits, the cue sentences and the published LM-rescored sentences of participant T15 for 1,718 trials over 14 days [2, 26]. We fit on the first half of each day (812 trials) and test on the second half (816 trials). We evaluate two decoders: our own phoneme-to-word route (CTC prefix beam over phonemes with a CMUdict lexicon and our word LM; WER 0.149) and

the published sentence (WER 0.043; 0.0434 pooled over all 1,718 trials), which we treat as a black box and gate.

Forearm sEMG (Toronto). The dataset holds isolated key presses from 19 participants over two days [8, 50]. Because it has no continuous typing, we compose test sentences from single presses of held-out Tatoeba sentences, with oracle press timing and no inter-key context (551 test sentences). A CNN classifies each press. Three conditions vary how much of the user's data the decoder saw: *session-calibrated* (day 1 plus half of day 2, cross-fitted), *cross-session* (day 1 only) and *cross-subject* (4-fold leave-participants-out).

5.4 Controls

To show that decoders use the neural signal, we replace the input with shuffled MEG, zeros, or train with permuted labels. To show that the trust layer does not commit hallucinated text, we feed it null streams: frame-shuffled streams, uniform streams, zero-input decoder output, a permuted-label decoder, and the real stream of a different sentence.

5.5 Statistics

We report 95% percentile bootstrap confidence intervals and paired differences on the same items. The resampling unit is stated with every interval: trials `[trial]` and days `{day}` for B2T '25; sentences `[sent]` and participants `{participant}` for sEMG. For the MEG trust results, the primary unit is the unique sentence (cluster bootstrap over the 17 test sentences, keeping all their instances; 2,000 resamples), with participant clusters as a second unit; intervals over instances ignore the variance across sentences and are too narrow. The MEG decoder CER intervals in Section 6.1 are still over instances. **[TODO: G3: sentence-level CIs for the MEG CER rows; day/session CIs for drift.]** Every decoder and trust run uses one seed. **[TODO: G2/G7: repeated seeds and repeated fit/test splits.]**

6 Results

6.1 The decoders use the neural signal

Before we evaluate the trust layer, we check that each decoder reads its input (Table 3). Our asynchronous MEG decoder reached a test CER of 0.506 with greedy decoding (0.495 ± 0.020 SEM over the 19 participants) and 0.475 with beam search and the character LM (Figure 5). With shuffled MEG the CER rose to 0.760 and with zero input to 0.925; the paired shuffled-minus-real gap was +0.254 [+0.241, +0.265] (instances). On the 11 test sentences with no close training variant, CER was 0.519 (greedy) and 0.492 (beam + LM), so near-variant sentences changed the headline by about 0.013 and 0.017. A longer training schedule did not help (r5 minus r4b, greedy +0.009 [+0.003, +0.015]). For reference, the published asynchronous encoder reports 0.59 ± 0.02 on SpanishBCBL [6]; this is not a head-to-head comparison, because the test splits differ and our decoder uses key times as training labels. On B2T '25, our phoneme-to-word route reached WER 0.149 and the published decoder 0.043. On sEMG, the CNN reached key accuracies of 0.299 (session-calibrated), 0.168 (cross-session) and 0.209 (cross-subject); with permuted training labels it fell to 0.022, the chance level.

6.2 The trust layer cuts wrong commits on three device types

Table 4 and Figures 1 and 6 give the main result.

Table 3: Decoder quality and input controls. CER/WER on the test sets; lower is better. MEG intervals are over instances. "Far" = the 11 MEG test sentences with no close training variant.

Device	Decoder	Real input	Controls
MEG	M1-09 async, greedy (CER)	0.506 (participants 0.495 ± 0.020)	shuffled 0.760, zeros 0.925
MEG	M1-09 async, beam + LM (CER)	0.475 (participants 0.462 ± 0.022)	—
MEG	far sentences, greedy / beam (CER)	0.519 / 0.492	—
MEG	v2 Encoder Async [6] (CER)	0.59 ± 0.02 (other split)	—
B2T '25	our phoneme-to-word route (WER)	0.149	—
B2T '25	published decoder (WER)	0.043 (0.0434 pooled, 1,718 trials)	—
sEMG	CNN, session-cal. / cross-session / cross-subject	key accuracy 0.299 / 0.168 / 0.209	permuted labels 0.022

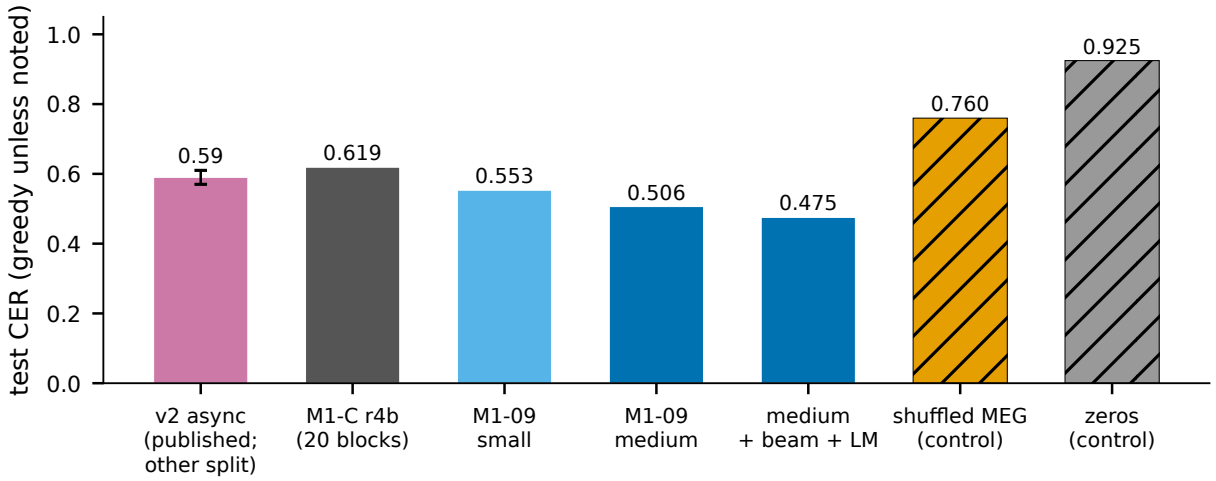


Figure 5: Our asynchronous MEG decoder uses the signal. Test CER on SpanishBCBL (637 instances, 17 unique sentences, 19 participants), mean over sentences. The published v2 value (mean ± SEM over participants) uses a different test split and is shown for orientation only. Hatched bars: input controls for the medium model (greedy). **[TODO: G2: add seed variability.]**

Intracortical speech. Always committing the top sentence of our decoder gave 11.45 eWPM and a wrong-commit rate of 0.610. The trust layer, on the same output, gave 11.55 eWPM and a wrong-commit rate of 0.059 [0.043, 0.075]. The paired eWPM difference was [-0.11, +0.30] and the wrong-commit difference [-0.587, -0.516] (trials). Used as a gate on the published LM-rescored sentences, the layer reduced the wrong-commit rate from 0.194 to 0.044 [0.031, 0.059] at a speed cost of 4.3% (paired eWPM [-0.98, -0.72]). All BCI policies were far faster than gaze typing alone (4.08 eWPM; trust minus gaze [+6.96, +7.97], gate minus gaze [+13.81, +15.53]).

Forearm sEMG. In all three decoder conditions the trust layer reduced the wrong-commit rate from 0.541, 0.882 and 0.962 to 0.004, 0.004 and 0.000, and it raised eWPM over always-commit (paired differences [+0.05, +0.38], [+0.44, +0.72] and [+0.43, +0.67], sentences). The layer committed only 0–4% of sentences; the speed gain came from offers, from which the user picks a candidate. Only the session-calibrated decoder clearly beat gaze typing (8.02 vs 4.27 eWPM; trust minus gaze [+3.38, +4.16]). The cross-subject condition stayed below gaze typing (4.08).

MEG. With our asynchronous decoder, the LM top sentence was right for 2 of 637 test instances (0.0031), and the true sentence was in the 10-best list for 1.6%. Always committing the top sentence therefore gave a wrong-commit rate of 0.997 and 2.580 eWPM [2.41, 2.76]. The trust layer committed nothing (wrong-commit 0) and offered candidates instead, which gave 3.203 eWPM [3.07, 3.36] (sentences; {3.08, 3.37} over participants), +0.62 over always-commit [+0.49, +0.75]. Gaze typing alone gave 3.155 [3.02, 3.29]. The paired difference to gaze typing

Table 4: Safety and speed on three device types (test sets). eWPM with 95% bootstrap CIs; square brackets: trials (B2T '25) or sentences (sEMG; MEG, cluster bootstrap over the 17 unique sentences); curly brackets: days (B2T '25) or participants (sEMG, MEG). The MEG wrong-commit CI is over sentences. Wrong-commit: share of items committed with an error. Silent error: wrong commit not noticed by the simulated user. "—": not computed.

Device / condition	Policy	eWPM [CI] {CI}	Wrong-commit	Silent
B2T '25, our decoder	always commit top-1	11.45 [10.95, 12.04] {10.72, 12.32}	0.610 [0.574, 0.645]	0.044
	trust	11.55 [11.03, 12.06] {10.85, 12.38}	0.059 [0.043, 0.075]	0.002
B2T '25, published	always commit	19.61 [18.68, 20.51] {17.76, 21.45}	0.194 [0.167, 0.222]	0.025
	trust gate	18.77 [17.88, 19.62] {17.05, 20.46}	0.044 [0.031, 0.059]	0.004
B2T '25	gaze typing only	4.08 [4.04, 4.12]	0	0
sEMG session-cal.	always commit LM top-1	7.81 [7.36, 8.30] {6.60, 9.01}	0.541	0.060
	trust	8.02 [7.63, 8.45] {6.90, 9.15}	0.004	0.000
sEMG cross-session	always commit LM top-1	4.10 [3.89, 4.34] {3.59, 4.70}	0.882	0.149
	trust	4.68 [4.52, 4.86] {4.21, 5.25}	0.004	0.000
sEMG cross-subject	always commit LM top-1	3.53 [3.37, 3.72]	0.962	0.167
	trust	4.08 [3.98, 4.20]	0.000	0.000
sEMG	gaze typing only	4.27	0	0
MEG (async M1-09)	always commit LM top-1	2.580 [2.41, 2.76] {2.41, 2.79}	0.997 [0.990, 1.000]	0.235
	trust	3.203 [3.07, 3.36] {3.08, 3.37}	0.000	0.000
MEG	gaze typing only	3.155 [3.02, 3.29]	0	0

was +0.05, with CIs of [-0.04, +0.15] over sentences and [-0.07, +0.22] over participants: this is parity, not a gain. Per participant, the trust layer was above gaze typing for 10 of 19 participants and below for 9 (range −0.34 to +1.16 eWPM). The synchronous M1-15 decoder with oracle key timing gave the same picture (trust 3.229, always-commit 2.714; trust minus gaze [-0.02, +0.18] over sentences).

6.3 Confidence predicts errors, and calibration helps

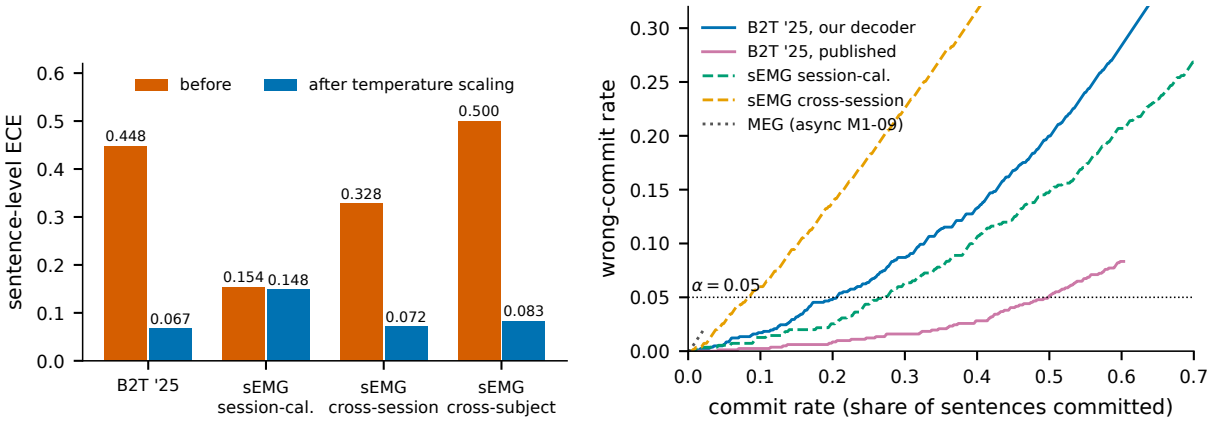
On B2T '25, the calibrated top-1 probability predicted a wrong top sentence with AUROC 0.797 at the word level and 0.884 at the phoneme level; per word, the acoustic word confidence gave AUROC 0.844. The raw logits were over-confident (frame temperature 3.78); sentence ECE fell from 0.448 to 0.067. On sEMG, AUROC was 0.83, 0.76 and 0.67 in the three conditions, and sentence ECE fell from 0.154 to 0.148, from 0.328 to 0.072 and from 0.500 to 0.083 (frame temperatures 1.25, 1.78 and 2.17) (Figure 7a). Figure 7b shows the trade-off between commit rate and wrong-commit rate as the commit threshold varies. The published B2T '25 decoder can commit about half of its sentences at the 0.05 level; our weaker decoder about a fifth; the cross-subject sEMG decoder produced no committable candidate. **[TODO: F5: reliability diagrams before/after need per-bin data that the committed JSON does not contain; add bins to the harness output.]**

6.4 Null input yields no committed text

Always-commit wrote a sentence for every null input: on B2T '25 for 100% of frame-shuffled and uniform streams, on sEMG for 100% of frame-shuffled, zero-input and permuted-decoder streams, and on MEG for 100% of shuffled-MEG, zero-MEG, frame-shuffled, uniform and other-sentence inputs, with fluent Spanish in 85–100% of them (85% on real input). The trust layer committed 0% of null inputs on all three devices (Figure 8). On B2T '25 and sEMG this is a real test, because the same layer did commit on real input (22% and 2.5% of the audited items). On MEG it is not: the layer committed 0% of real inputs as well, so the MEG audit shows that the gate stayed closed, not that it can tell null input from real input. One control it cannot pass: given the *real* stream of another sentence, the trust layer on our B2T decoder committed



Figure 6: Speed per policy and device. eWPM of the simulated user; orange: always commit, blue: trust layer. Error bars: 95% bootstrap CIs over trials (B2T '25) and sentences (sEMG; MEG, 17 unique sentences). Dashed lines: gaze typing alone. The trust layer never made the system slower than always-commit except for the published-decoder gate (−4.3%).



(a) Sentence-level ECE of $\hat{p}_{\text{top1}}$.

(b) Wrong-commit rate vs commit rate.

Figure 7: Calibration and the commit trade-off. (a) Expected calibration error of the sentence confidence before and after temperature scaling (test sets). (b) For each commit threshold on the calibrated score, the share of all test items that are committed (x) and committed wrongly (y); items blocked by the guard or other policy rules are never committed. Dotted line: $\alpha = 0.05$. The cross-subject sEMG condition is absent because no candidate passed the guard.

22% of items, the same rate as for real input, and on sEMG it committed as often as for real input. The published-sentence gate committed 0% in this case, because the published text did not match the other sentence's stream. A layer that sees only the stream cannot detect that the user intended a different sentence.

6.5 Conformal risk control fixes the mean wrong-commit rate

The plug-in policy exceeded the target on B2T '25 with our decoder (0.059 on the full test set). Over 200 random calibration/test splits with 408 calibration trials (Table 5, Figure 9), CRC brought the mean test wrong-commit rate to 0.0483 (our decoder) and 0.0478 (published gate), at an eWPM change of −0.3% and +0.1%. LTT gave 0.0345 for both. CRC controls the mean, not every split: 41–45% of CRC splits were above 0.05 and the 95th percentile was 0.071–0.076. LTT lowered the 95th percentile to 0.056–0.061. On MEG, where almost every possible commit is wrong, CRC still committed 2.1% of trials (wrong-commit 0.0193; LTT 0.0214) because this is below the budget; it gained no speed (eWPM 3.199 vs 3.204 for the plug-in gate, which committed nothing). Calibration splits grouped by sentence (11 of 17) or by participant (12 of

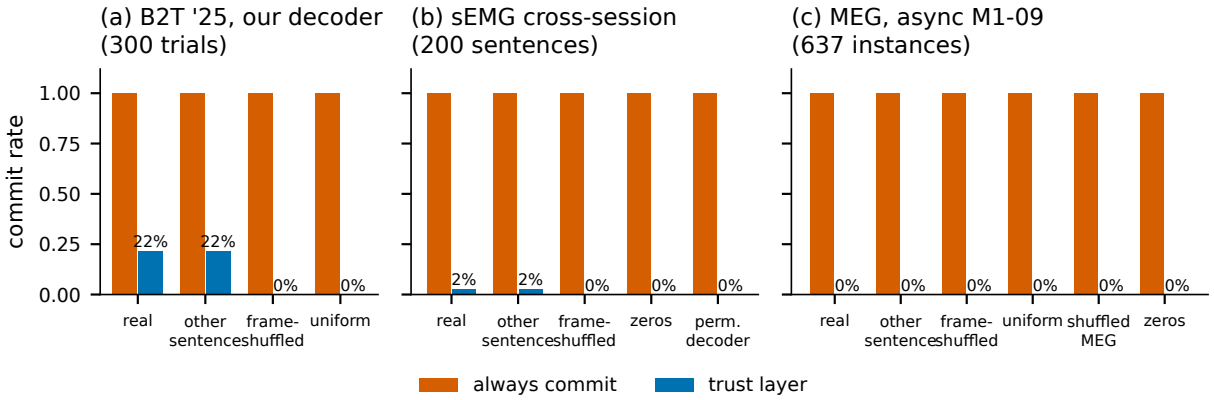


Figure 8: Null-input audit. Commit rate of always-commit (orange) and the trust layer (blue) on real and null inputs. (a) B2T '25, our decoder, 300 random test trials. (b) sEMG cross-session, 200 random test sentences. (c) MEG, asynchronous M1-09 decoder, all 637 test instances; the trust layer also committed 0% of real inputs, so this panel shows a closed gate, not null detection. For "real" and "other sentence", commits are a mix of right and wrong; for all other inputs any commit is wrong.

Table 5: CRC and LTT for the commit threshold ($\alpha = 0.05$, $\delta = 0.1$, 408 calibration trials, 200 random splits). Mean over splits; p95: 95th percentile over splits. MEG: async M1-09 streams.

Data	Method	Wrong-commit	p95	Commit rate	eWPM
B2T '25, our decoder	plug-in gate	0.0586	0.071	0.233	11.57
	CRC	0.0483	0.071	0.194	11.53
	LTT	0.0345	0.056	0.156	11.52
B2T '25, published gate	plug-in gate	0.0441	0.059	0.471	18.77
	CRC	0.0478	0.076	0.479	18.78
	LTT	0.0345	0.061	0.427	18.70
MEG (async M1-09)	plug-in gate	0.0000	0.0000	0.000	3.204
	CRC	0.0193	0.0306	0.021	3.199
	LTT	0.0214	0.0349	0.023	3.199

19) gave the same picture (CRC 0.0194 and 0.0201). On sEMG, with 142 calibration sentences, both methods were feasible (cross-session CRC threshold 0.204, bound 0.049) and equal to the stricter plug-in policy.

6.6 Online re-calibration restores the target under drift

CRC assumes exchangeable items. When we calibrated on the earlier half of each B2T '25 day and applied the threshold to the later half in time order, offline CRC gave 0.0625; on sEMG cross-session it gave 0.0635 (Table 6). The online methods brought both to 0.040–0.053 at an eWPM cost of at most 0.6%. `aci_window` stayed below 0.05 in every drift case (0.0478 and 0.0490) and `aci` sat at the target (0.0502 and 0.0526), as its bound predicts. Figure 10a shows the running rate. Feedback delayed by 10 trials changed little (0.0404–0.0502 on B2T '25, our decoder). Without a calibration block (cold start over all 1,628 trials), `aci` gave 0.0442 and `aci_window` 0.0491. No method controlled short windows: the maximum over 50-trial windows was 0.10–0.12 for all methods, as for an i.i.d. Bernoulli(0.05) sequence (median 0.120). When only noticed errors were fed back, `aci` gave 0.0515 on B2T '25 and 0.0635 on sEMG cross-session, because silent errors look correct to the controller.

Two CRC numbers on sEMG. Section 6.5 reports a wrong-commit rate of 0.004 for CRC on sEMG cross-session, and Table 6 reports 0.0635. Both are correct. The first combines CRC

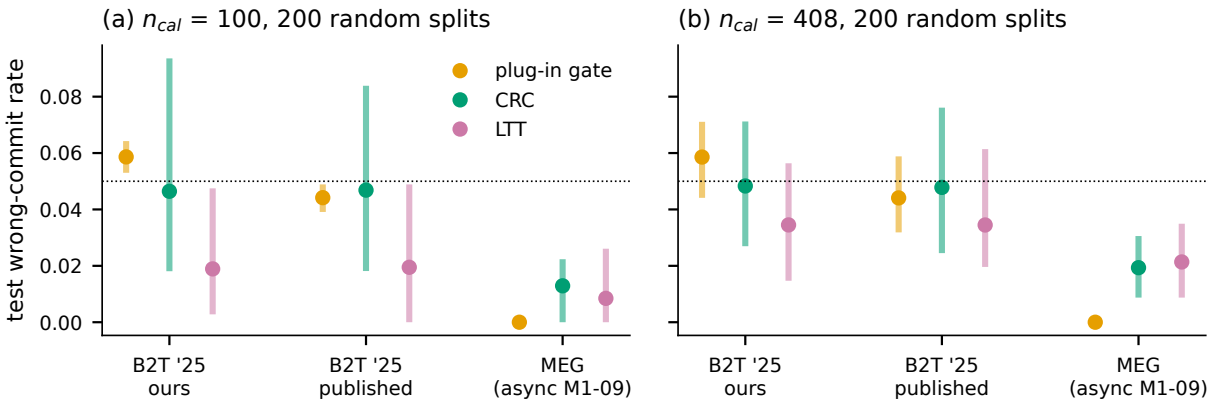


Figure 9: CRC controls the mean, LTT also the tail. Test wrong-commit rate over 200 random calibration/test splits with (a) 100 and (b) 408 calibration items. Dots: mean; bars: 5th to 95th percentile. Dotted line: $\alpha = 0.05$.

Table 6: Wrong-commit rate under within-session drift. Warm start on earlier data, then the later data in time order. eWPM in brackets. MEG: async M1-09 streams in participant order, warm start on validation data; almost every commit is wrong. **[TODO: G3: CIs over days/sessions.]**

Data	Offline CRC	`window`	`decay`	`aci`	`aci_window`
B2T '25, ours	0.0625 (11.553)	0.0453 (11.490)	0.0417 (11.500)	0.0502 (11.510)	0.0478 (11.489)
B2T '25, published	0.0490 (18.80)	0.0429 (18.77)	0.0392 (18.74)	0.0478 (18.80)	0.0453 (18.77)
sEMG cross-session	0.0635 (4.657)	0.0508 (4.669)	0.0399 (4.670)	0.0526 (4.667)	0.0490 (4.668)
MEG (async M1-09)	0.0204 (3.198)	0.0078 (3.201)	0.0204 (3.198)	0.0188 (3.200)	0.0078 (3.201)

with the stricter eWPM-tuned threshold ($\max(\lambda_{\text{tuned}}, \hat{\lambda}_{\text{CRC}})$); the second uses the pure CRC threshold, which exposes the drift that the stricter threshold hid.

6.7 The ledger traces every wrong commit

We replayed the B2T '25 test half (816 trials) with the ledger attached, for the plug-in gate and a CRC gate. The ledger held 1,723 entries and passed verification. It found the same wrong commits as the harness: 48 of 48 (plug-in) and 39 of 39 (CRC), of which 2 and 1 were silent. Every tampering case we tried was detected: editing an entry, editing with re-hashing, deleting, reordering, and cutting the tail (with the head anchor). The cost was about 0.1 ms and about 1.1 kB per decision. Figure 11 shows one entry of the audit report: most wrong commits were one-word errors.

6.8 Ablations

On B2T '25, removing calibration raised the wrong-commit rate from 0.059 to 0.071 (eWPM 11.57 vs 11.55), and removing the guard lowered it to 0.036 (eWPM 11.51): the guard did not help on this device. On sEMG session-calibrated, removing calibration raised the wrong-commit rate from 0.004 to 0.045 and eWPM from 8.02 to 8.15. **[TODO: G5: one unified ablation table over the three devices (calibration, guard, conformal sets, offer_words, CRC on/off), same seeds.]**

7 Discussion

A trust layer makes a fixed decoder safer, not faster. On every device type, the same layer, with only a new adapter, turned frequent wrong commits into rare ones. Where the

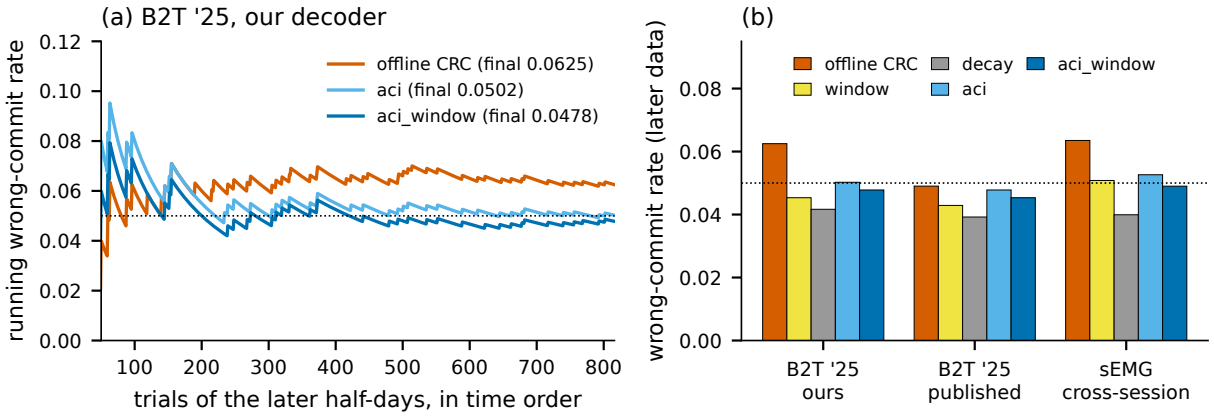


Figure 10: Online re-calibration keeps the long-run rate at the target under drift. (a) Running wrong-commit rate over the later half-days of B2T '25 (our decoder, 816 trials, time order), recomputed from the committed per-trial items; final values equal Table 6. The first 50 trials are not shown. (b) Final wrong-commit rate per method and dataset. Dotted line: $\alpha = 0.05$.

Wrong commit `dec-000041-ed60b4bd` (entry 44, hash `8cf2c6c67053b185...`)
Committed: "who you never expect to when"
Correction (oracle ground truth, hash-linked): "who you never expect to win"; 2 character edits, 1 word edit
Scores: $\hat{p}_{\text{top1}}$ 0.5814, set size 3, guard risk 0.2857; reason: $\hat{p}_{\text{top1}} \geq 0.50$
Input: stream `1b4b60e8e4950710...`, frames [0, 335], item `b2t25-test-trial-196`
Decoder: published online RNN + our phoneme-to-word decoder; LM: Tatoeba character 6-gram + word unigram + CMUdict; adapter `fidetype.adapters.b2t25`
Calibration `cal-692fb52a73417552`: T_f 3.7809, T_s 4.5551, fit set: first half of each day (812 trials)
Policy `pol-2eaa4ff51919527a`: commit threshold 0.5, maximum guard risk 0.5, offer size 2

Figure 11: One entry of the provenance audit report (B2T '25 replay, plug-in gate; public CC0 cue sentence). Each wrong commit is traced to the exact stream segment and to the decoder, LM, calibration and policy versions that produced it. Hashes are shortened.

decoder was good enough for its top sentence to be right often (intracortical speech, session-calibrated sEMG), this came at no speed cost and the system stayed far faster than gaze typing. Where the decoder was weak (MEG, cross-subject sEMG), the layer refused to commit and fell back to offers or typing, so the system stayed at or below gaze typing. The layer cannot create information that the decoder does not have; it can only stop the LM from presenting a guess as an answer. On MEG, our result is therefore a safety result, not a usefulness result.

Strong decoders still need a gate. The published B2T '25 decoder has a WER of 0.043, yet it committed a wrong sentence for 19% of trials. Gating its text by agreement with the raw neural evidence kept 96% of its speed and reduced wrong commits to 4%. This gate needed only the stream and the text, not the decoder's weights, which is the use case a vendor-neutral format is meant to enable.

Per-session calibration matters more than the gate for speed. On sEMG, a decoder that saw half a session of the user's data roughly doubled eWPM over one trained on another day (8.02 vs 4.68). The trust layer made both safe; only calibration made one fast. For a product, a short calibration block at the start of each session is worth more than any downstream policy, and the same block can serve as the calibration set for CRC.

How to state the guarantees. Our results suggest a precise wording. With CRC: "the expected wrong-commit rate is at most 0.05 for this user in this session, given n calibration

sentences, if the session does not drift"; individual sessions can exceed it. With LTT: "with probability at least 0.9, the wrong-commit rate is at most 0.05". With online re-calibration: "the average wrong-commit rate over the T items of this session is at most 0.05 plus a term of order $1/T$, for errors the system observes". Each statement should come with the measured silent-error rate, because silent errors are outside all of them.

What standardisation buys. One stream format let us apply one gate, one risk controller and one ledger to three device types with a few hundred lines of adapter code each. The conformance suite turned implicit assumptions into checks and found real bugs in our own producers. We see this as the first step towards a shared contract between decoder vendors and the applications that consume their output, in the way OpenTelemetry separated instrumentation from analysis back-ends. **[TODO: decide whether to keep the OpenTelemetry analogy for the target venue.]**

8 Limitations

1. **One implant participant.** B2T '25 contains participant T15 only, in a copy task, with online logits that we cannot re-run. Our intervals cover trials and days, not people.

2. **Few unique MEG test sentences.** The MEG test set has 637 instances but only 17 unique sentences. Our MEG trust intervals use a cluster bootstrap over these 17 sentences, which is still a small number of clusters; the decoder CER intervals are over instances and understate the variance across sentences. Six of the 17 test sentences have a close training variant (about 0.013 greedy CER effect); we report the far-sentence numbers next to the headline.

3. **No head-to-head MEG comparison.** Our MEG test split differs from that of Brain2Qwerty v2, and our decoder uses key times as training labels (not as inputs). The comparison of 0.495 with 0.59 is not a head-to-head result, and we do not claim a better decoder.

4. **MEG reaches only parity with gaze typing** (trust minus gaze +0.05, 95% CI over sentences $[-0.04, +0.15]$). The true sentence is in the 10-best list for 1.6% of test instances, so the trust layer commits nothing on MEG; the MEG results show safety, not usefulness, and the MEG null audit shows a closed gate rather than null detection.

5. **Composed sEMG sentences.** The Toronto data has isolated key presses only; sentences are composed with oracle timing and without inter-key context. Natural typing is untested. **[TODO: G4: natural-typing sEMG (KT-sEMG) or keep as limitation.]**

6. **No live users.** All eWPM values come from a simulated user. Only the fallback typing speed and the selection time are matched to measurements [16]; the probability of missing an unflagged error has no measurement. With a 40 characters-per-minute fallback, gaze typing becomes clearly faster than the MEG system (6.31 vs 5.15 eWPM). **[TODO: G14: user study (needs ethics approval).]**

7. **Silent errors are outside every guarantee.** With noticed-only feedback, `aci` gave 0.0515 (B2T '25) and 0.0635 (sEMG).

8. **Guarantees are marginal or long-run.** CRC bounds the mean over sessions (41–45% of splits exceeded 0.05); online bounds are long-run averages; 50-trial windows reached 0.10–0.12.

9. **Stream-only gating cannot detect intent mismatch.** A real stream of another sentence was committed as often as real input.

10. **One seed and one split.** Every decoder and harness run used one seed, and the B2T '25 and sEMG trust runs one fit/test split. **[TODO: G2, G7.]**

11. **Simple language models.** We use a character 6-gram and a word unigram trained on Tatoeba. 64 of the 1,628 B2T '25 cue sentences occur in Tatoeba English (a possible small leak that we did not remove). **[TODO: G9: rebuild the LM without them and rerun.]**

12. **The ledger is tamper-evident, not tamper-proof**; it has no signature, and in live use a wrong commit is known only after a correction.

9 Ethics and data licences

This study uses only public, de-identified datasets. No new data were collected and no experiment with human participants was run, so no new ethics approval was required. **[TODO: the user decides the final wording of this statement; check the venue's policy on secondary use of public data.]** We did not attempt re-identification. Raw recordings and participant data are not redistributed; our repository contains code, configurations, derived summary statistics and audit logs only. Table 7 lists every dataset and resource and how we use it.

Table 7: Datasets, resources and licences.

Dataset / resource	Licence	Use	Note
SpanishBCBL [5, 48]	CC BY-NC 4.0	MEG decoder training and evaluation	Research only; weights trained on it are not released for commercial use
Brain-to-Text '25 copy-task logits [2, 26]	CC0 1.0	intracortical trust evaluation	Companion code has no licence; we wrote our own loader
Toronto sEMG typing [8, 50]	CC BY 4.0	sEMG decoder and trust evaluation	Baseline code (CC BY-SA 4.0) not used
Tatoeba English / Spanish sentences	CC BY 2.0 FR (some CC0)	character LMs; composed sEMG sentences	Sentences from Tatoeba (`https://tatoeba.org`), by the Tatoeba contributors
CMUdict (commit 7479086)	BSD-2-Clause style	phoneme-to-word lexicon	Copyright notice kept
Brain2Qwerty code (Meta)	CC BY-NC 4.0	read only, not copied	Our decoders are our own implementations
Fidetype code and specification	Apache-2.0	all methods	—

10 Code and data availability

The specification (NeuralTextStream v0.1), the conformance suite with its golden vectors, the reference implementation (`fidetype`: adapters, decoders, trust layer, online risk control, provenance ledger) and all experiment scripts and configurations are available under Apache-2.0 at `https://github.com/zhangzibo/fidetype`. Every table and figure in this paper is generated from committed result files; the figure script is `papers/paper1/make_figures.py`. The datasets

are available from their original sources (Table 7); each has a fetch note with a SHA-256 digest in the repository. **[TODO: tag the commit used for the paper (`paper1-v1`) and archive it (e.g. Zenodo DOI) before submission.]** Hardware: one AMD RX 7800 XT GPU (ROCm) for decoder training; all trust-layer runs use a CPU. **[TODO: G12: report trust-layer run time per decision (only the ledger cost is measured).]**

11 Conclusion

We introduced an open stream format for neural decoder output and a trust layer that reads only this format. On three device types with public data, the same layer reduced wrong commits from 0.19–1.0 to at most 0.059, committed nothing from null input, controlled the mean wrong-commit rate at the target with conformal risk control, and kept it there under drift with online re-calibration, while a hash-chained ledger traced every wrong commit. Where the decoder was good, safety came at no speed cost; where it was weak, the layer made the system safe but not useful. Next steps are a study with gaze-typing users, which needs ethics approval; a streaming MEG decoder; a version 1.0 of the specification with third-party conformance; and a public benchmark built on the same harness and controls.

Acknowledgements

[TODO: funding, if any.] The authors used AI coding and writing assistants under human review for code, experiment scripts and drafting; all numbers were generated by committed scripts and checked against the committed result files. **[TODO: adjust this statement to the policy of the chosen venue.]**

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A User model parameters

Table 8 lists the parameters of the simulated user (`UserConfig.aac_gaze()` in module `fidetype.interact.simulated_user`). Only the fallback typing speed and the sum of the per-slot selection time are matched to measurements [16]; all other values are assumptions.

Table 8: Simulated-user parameters.

Parameter	Value	Basis
Fallback (gaze keyboard) speed	20 characters/min	measured, eye-gaze user LP1 [16]
Per-slot scan + select ($k = 3$)	$1.0 + 1.0 \log_2(k+1) + 1.5 = 4.5$ s	sum matched to 4.7 s [16]
One gaze selection of a highlighted target	1.5 s	assumption
Navigate to a wrong word	3.0 s (2 selections)	assumption
Reading time	0.3 s per word, 0.5 s per candidate	assumption
Probability of noticing a wrong commit	0.9	assumption
Miss probability of an unflagged wrong word	0.05	assumption (no measurement)
Correction overhead	2.0 s	assumption
Neural input speed, MEG / B2T '25 / sEMG	152 / 159.3 / 75.8 characters/min	device setting

[TODO: G6: sensitivity of eWPM and of the policy ranking to these values on all three devices.]

B MEG decoder and leakage checks

The M1-09 decoder is a convolutional front end (320 channels) followed by 4 Conformer layers of width 256 [49], trained with CTC [27] and a keystroke auxiliary loss that uses key times as training labels (stage r4a pretraining, then r4b on all 80 cached blocks). At inference, the forward pass `ConvCTC.forward(x, lengths, subjects)` takes the MEG segment, its length and the participant index; it has no key input. Table 9 compares the inference inputs with the published asynchronous decoder.

Table 9: Asynchronous setting: our decoder vs v2.

Input at inference	M1-09 (ours)	v2 async [6]
MEG of the sentence segment	yes	yes
Segment start	first key − 0.4 s	first key − 400 ms
Segment end	last key press + 0.5 s	last key release + 400–500 ms
Per-key times or key count	no	no
Participant index	yes	yes
Key times as training labels	yes (auxiliary loss)	no

Leakage checks: the split is by a hash of the sentence text, and no test sentence occurs in the LM training text. On the synchronous M1-15 decoder we found no exact leakage and

a near-variant memorisation effect of about 0.013 CER. With the synchronous M1-15 decoder (oracle key timing; greedy CER 0.508, beam + LM 0.447), the trust results were the same as with the asynchronous decoder: always-commit 2.714, trust 3.229 and gaze typing 3.155 eWPM, trust minus gaze $[-0.02, +0.18]$ over sentences. Both decoders take the sentence bounds from the keyboard log, so neither is a free-running, real-time decoder. **[TODO: G8: k-fold over sentence hashes to increase the number of unique test sentences.]**

C Additional results

[TODO: Appendix D: full CRC/LTT tables at 100 and 408 calibration items and online hyper-parameter sensitivity (η grid). Appendix E: simulator results (supplementary only). Appendix F: provenance schema and a full audit report.]